INCH-POUND

MIL-P-70624A (AR)
AMENDMENT 3
24 March 1992

SUPERSEDING
AMENDMENT 2
10 April 1989

MILITARY SPECIFICATION

Power Box Components Assembly, Electronics Box Components Assembly, Cable Assemblies and Transmitter Assembly for Mine Dispenser: M138 (FLIPPER)

This amendment forms a part of Military Specification MIL-P-70624A (AR), dated 15 May 1987, and is approved for use by the US Army Armament, Munitions and Chemical Command) and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

* 2.1.1 Specifications and Standards, add the following under SPECIFICATIONS:

"FEDERAL"

TT-C-490 - Cleaning Methods and Pretreatment of Ferrous Surfaces for Organic Coatings."

PAGE 2

* 2.1.1 Specifications and standards, delete MIL-S-45743. MMIL-C-46168, and MIL-S-46844 in their entirety.

PAGE 3

* 2.1.1 Specification and standards, add the following standards:

"MIL-STD-202 Test Methods for Electronic and Electrical Component Parts".

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:49Z
Check the source to verify that this is the current version before use.

MIL-STD-2000, Standard Requirements for Soldered Electrical and Electronic Assemblies."

PAGE 5

* 3.4.1 Power output table:

Delete "E-Less than 1" and substitute "E-Less than 2".
Delete "K-11 ± 1" and substitute K-12.5 + 1".
Delete "V-6 ± .1" and substitute 'V-6.9 ± .4".
Delete "Y-11 ± 1" and substitute "Y- 12.5 ± 1".

* 3.4.2, Power output table:

Delete "E-Less than 1" and substitute "E-Less than 1.8".
Delete "U-14 + 1" and substitute "U- 13 + 1".
Delete "X-5.95 ± .1" and substitute "X-6.6 ± .3".

PAGE 6

* 3.5.1 Power output table:

Delete "B-11 ± 1" and substitute "B-13 ± 1".
Delete "C-13 ± 1" and substitute "C-14 ± 1".
Delete "D-13 ± 1" and substitute "D-14 ± 1".
Delete "H-44 ± 1" and substitute "H-45 ± 1".
Delete "K-13 ± 1" and substitute "K-14 ± 1".
Delete "T-21.5 ± 1" and substitute "T-2.9 ± 1.3".

* 3.5.2 Power output table:

Delete "A,P - 0 + .1" and substitute "A,P - 0 ± .1".
Delete "B- 11 + 1" and substitute "B- 13 ± 1".
Delete "C- 13 ± 1" and substitute "C- 14 ± 1".
Delete "D- 13 ± 1" and substitute "D- 14 ± 1".
Delete "H- 44 ± 1" and substitute "H- 45 ± 1".
Delete "K- 13 ± 1" and substitute "K- 14 ± 1".
Delete "T- 21.5 ± 1" and substitute "T- 21.9 ± 1.3".

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:49Z
Check the source to verify that this is the current version before use.

PAGE 7

3.6.3 Electronics box assembly, power output table:
Delete voltage value "13 ± 1" and substitute "14 ± 1".

* 3.6.4 Electronics box assembly, power output table:

Delete voltage value #4, "21 ± 1" and substitute "21.9 ± 1.3".
Delete voltage value #5, "10 ± 1" and substitute "12.5 ± 1".
Delete voltage value #6, "13 ± 1" and substitute "14 ± 1".
Delete voltage value #7, "Less than 1 volt" and substitute "Less than 1.8 volts".
Delete Pin and Voltage values #8 and substitute:

"Pin	Voltage
P7-2(-) and P7-3(+)	26 ± .5"

Delete Pin and Voltage values #9 and substitute:

"Pin	Voltage
TB3-1(-) and TB3-10(+)	26 ± .5"

Delete Pin and Voltage values #10 and substitute:

"Pin	Voltage
TB3-1(-1) and TB3-11(+)	4 ± 1 AC"

Acid Pin and Voltage values #11:

"Pin	Voltage
TB3-1 (-) and TB3-12(+)	5.2 to 5.4 volts @ trigger point (for information only)"

Source: https://assist.dla.mil -- Downloaded: 202610-23T03:49Z
Check the source to verify that this is the current version before use.

PAGE 8

* 3.6.9 Electronics box assembly, power output table:

Delete voltage value #4, "21 ± 1" and substitute "21.9 ± 1.3".
Delete voltage value #6, "13 ± 1" and substitute "14 ± 1".
Delete voltage value #7, "Less than 1 volt", and substitute "Less than 1.8 volts".
Delete Pin and Voltage values #8 and substitute:

"Pin	Voltage
P7-(-) and P7-3 (+)	26 ± .5"

Delete Pin and Voltage values #9 and substitute:

"Pin	Voltage
TB3-1 and TB3-10	26 ± 1"

Delete Pin and Voltage values #10 and substitute:

"Pin	Voltage
TB3-1(-) and T33-11 (+)	4 ± 1 AC"

Add Pin and Voltage values #11:

'Pin	Voltage
TB3-1(-) and TB3-12(+)	5.2 to 5.4 volts @ trigger point (for information only)"

PAGE 9

* 3.6.11 Electronics box assembly, line 3: delete; "...between 6 to 10 seconds and substitute:

* . ..between 3 to 9 seconds".

PAGE 10

3.7.1 Welding, Change the second sentence to read:

"All welds shall be classified Class A, unless otherwise specified".

Source: https://assist.dla.mil -- Downloaded: 202610-23T03:49Z
Check the source to verify that this is the current version before use.

* 3.7 Procedures, add the following paragraph:

"3. 7.3 Protective finish. The contractor shall prepare two(2) test panels of the same material as used in components at the time at finishing the components or assemblies in accordance with MIL-STD-171, Green 383, subjected to the same finishing process and shall meet the requirements of TT-C-490 when tested as specified in 4.5.5.

PAGE 23

* 4.4.2.8 Body, Power Box, Category 119, delete footnote indicator, "1/" and delete '4.5.5" paragraph reference.

PAGE 35

* 4.4.2.17 Power Box Components Assembly, add footnote indicator "1/", to Categories 104 and 105 in the No. of Sample Units column. Add the following footnote in the Notes block:

"1/" Cold and hot temperature testing are required only on units which will be provided as spares, per 4.5.1."

PAGE 38

* 4.4.2.20 Terminal Marker TB1: Electronic Box, Category 103, delete "Gage" and substitute "Visual".

PAGE 45

* 4.4.2.24 Electronic Box, Categories 134 and 135, delete footnote indicators, "1/", and delete "4.5.5" paragraph reference.

PAGE 46

* 4.4.2.25 Cover, Access: Electronics, Category 103, delete footnote indicator, "1/", and delete "4.5.5" paragraph reference.

PAGE 48

* 4.4.2.27 Panel, Electronics BOX, Category 133, delete footnote indicator, "1/" and delete "4.5.5" paragraph reference.

PAGE 50

* 4.4.2.29 Cover Access: Test Block, category 106, delete footnote indicated, "1/", and delete "4.5.5" paragraph reference.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:49Z
Check the source to verify that this is the current version before use.

PAGE 53

* 4.4.2.32 Resistor, Fixed, Composition (Insulated), Part Number RCR07G 68 OJS, delete part Value "68 K OHM" and substitute "68 OHM".

PAGE 55

* 4.4.2.34 Resistor, Fixed, Film, Established, Reliability Style RNR65 or RNC65, delete:

"Part number RNC65H1582FS - Part value 15.8 K OHM" and substitute:

Part number RNC65H1782FS - Part value 17.8K OHM"

PAGE 62

* 4.4.2.41 Capacitor, Fixed, Electrolyte, Category 104, delete in its entirety,

PAGE 63

* 4.4.2.42 Semiconductor Device, Diode, delete the following:

Part Number JANTX 1N3021B - Part Value 11 volts" and substitute:

"Part Number JANTX 1N3023B - Part Value 13 Volts".

PAGE 72

4.4.2.51 Major 102, Under Inspection Method: Delete "DC" and substitute "AC".

PAGE 73

4.4.2.52 For Major 101 and 102: Change "AQL or 100%" column entry:

From: "100"
To: "0.40%"

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:49Z
Check the source to verify that this is the current version before use.

PAGE 78

* 4.4.2.55 Electronics Components Ass'y, add footnote indicator "1/" to Categories 118 and 119 in the No. of Sample Units column. Add the following footnote in the Notes block:

"1/ Cold and hot temperature testing are required only on units which will be provided as spares, per 4.5.4".

PAGE 79

* 4.5.1 Power box components assembly (dwg. 9359910), add the following:

"NOTE: The cold and hot temperature testing of the Power Box Components Assembly, as specified in 4.5.1.4 and 4.5.1.5 are not required to be performed if the Power Box Components Assembly will be cold and hot temperature tested as an integral part of the FLIPPER Mine Dispenser (MIL-D-70631). Cold and hot temperature testing are required, therefore, only on units which will be provided as spares".

4.5.1.4, line 3, change from "Method 502.1" to "Method 502.2".

PAGE 80

* 4.5.2.2 Circuit board assembly test procedure, b. change from "1000 HZ" to "1445 HZ".

Delete paragraphs through 'i" "c" in their entirety and substitute:

"c. Apply board power per para. 4.5.2.1 and on Pin Z, apply the 1445-HZ 10V P-P Sinewave.

d. Turn the potentiometer R147 counterclockwise at the starting point.

e. Connect the positive lead of the multimeter to pin B. Adjust potentiometer R147 clockwise so that the voltage at pin B is 5.4 V ± .1 V.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:49Z
Check the source to verify that this is the current version before use.

f. Connect the positive lead of the multimeter to U108-1. Adjust potentiometer R147 clockwise slowly until the voltage at U108-1 is steady above 6.5V. Turn the pot counterclockwise slowly until the voltage at U108-1 just reads less than 1 volt. Turn the potentiometer clockwise slowly until the voltage of U108-1 just rises and reads steady above 6.5 volts.

g. Decrease the frequency slowly down to 1440 Hz. The voltage at pin U108-1 should be less than 1 volt.

h. Increase the frequency slowly to 1450 Hz. The voltage at pin U108-1 should read steady above 6.5 volts.

i. Connect the positive lead of the multimeter to pin B. The meter should read between 5.30 and 5.70 volts. Turn power and signal. generator off."

PAGE 81

* Figure 3, for switch S1 in the normal position,

Delete: "11 ± .5V"
Substitute: "13 ± .65V".

PAGE 82

* 4.5.2.4 Circuit board assembly test procedure, delete subparagraph "a" in its entirety and substitute:

"a. Turn on the board power per para. 4.5.2.1, the signal generator and power to switches S1 and S2 (as shown in Figure 3). Set switch S1 to normal. Voltage at Pin E should be 13 ± 1 volts the first 3-9 seconds and after that should read less than 1.8 volts. Pin U should read 13 ± 1 volts after switch S2 is pressed momentarily".

* 4,5.2.5 Circuit board assembly test procedure, subparagraph "a", delete first sentence in its entirety. Delete subparagraph "b" in its entirety and substitute:

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:49Z
Check the source to verify that this is the current version before use.

"b. Activate switch S3 momentarily. Pin E should read 13 ± 1 volts the first 3 to 9 seconds and after that should read less than 1.8 volts. Scope reading should be 7 ± 2 volts peak decaying to 0 volts over 6 milliseconds. (Figure 4) Turn off oscilloscope".

Subparagraph "c", second sentence, delete '1 volt" and substitute "1.8 volts".

Subparagraph "d", fourth sentence, delete '.. .less than 1 Volt". and substitute ".. .less than 1.8 volts".

PAGE 83

* 4.5.2.6 Circuit board assembly test procedure, subparagraph "a", delete first sentence in its entirety. "Delete subparagraph "b" in its entirety and substitute:

"b. Activate S3 switch. Pin E should read 13 ± 1V the first 3 to 9 seconds and less than 1.8 volts after that. scope reading should be 29 ± 3 volts peak decaying to 0 volts over 6 milliseconds. (Figure 5)".

Subparagraph 'c", second sentence, delete "1 volt" and substitute "1.8 volts'.

Subparagraph "e", fourth sentence, delete ".. .less than 1 volt' and substitute '.. less than 1.8 volts". "

4.5.3.1 Circuit board assembly test procedure, Figure 6, delete "+9 VDC" and substitute "+9 ± .5 VDC".

PAGE 85

* 4.5.4 Electronics box components ass'y test procedure, add the following note before existing note:

"NOTE 1: The cold and hot temperature testing of the Electronics Box Components Assembly, as specified in 4.5.4.15 and 4.5.4.16, are not required to be performed if the Electronics Box Components Assembly will be cold and hot temperature tested as an integral part of the FLIPPER Mine Dispenser (MIL-D-70631). Cold and hot temperature testing are required, therefore, only on units which will be provided as spares". Delete existing "NOTE" and substitute "NOTE 2".

* 4.5.4.4 Circuit board assembly test procedure, Add the following sentence after third sentence:

"Install a jumper from a pin P7-4 to pin P7-5".

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:49Z
Check the source to verify that this is the current version before use.

PAGE 86

* 4.5.4.8 Circuit board assembly test procedure, delete first sentence in its entirety.

* 4.5.4.9 Circuit board assembly test procedure, delete subparagraphs "b" and "C" In their entirety and substitute the following:

'b. Set generator to 1200 HZ. Turn the mode switch to 'extended position". Increase frequency of sine wave generator slowly until ready light is on. use a multimeter to measure and observe the requirements specified in 3.6.9".

PAGE 87

* 4.5.4.11 Circuit board assembly test procedure, subparagraph 'b", add the following:

"Figure 7 shall be supplied by an open collector network simulating the opening and closing of a switch within the time specified by Figure 7. The open collector shall be connected to TB3-2 which supplies approximately 21VDC".

* 4.5.4.11 Delete Figure 7 in its entirety and replace with the following:

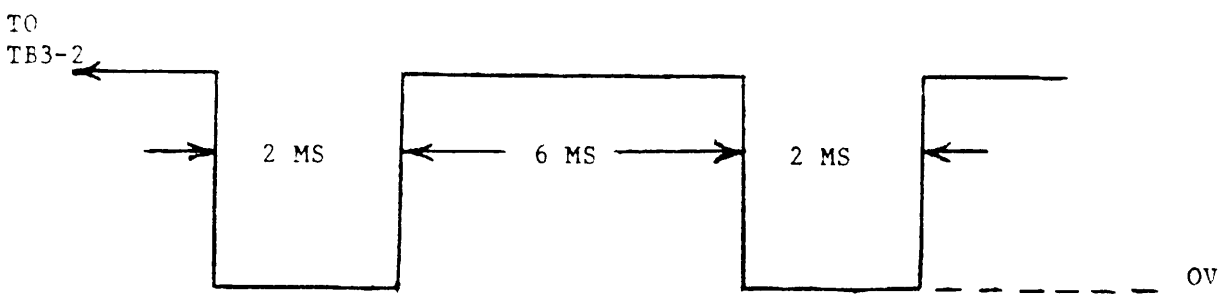


FIGURE 7

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:49Z
Check the source to verify that this is the current version before use.

* 4.5.4.13 Circuit board assembly test procedure, delete third sentence in its entirety and substitute:

"Disconnect the power supply, the jumper from P7-4 to P7-5, the proximity switch, cable and the transmitter coil assembly".

PAGE 88

* 4.5.5 Protective finish, delete paragraph in its entirety and substitute:

"4.5.5 Protective finish. The specimens shall be tested as specified in TT-C-490, 4.2.8 and 4.2.9 for the requirements in 3.7.3. The panel shall be exposed to salt spray for 48 hours minimum" .

The margins of this amendment are marked with an asterisk or vertical line to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1345-A604)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:49Z
Check the source to verify that this is the current version before use.